Linking Artificial Intelligence and Supply Chain Resilience: Roles of Dynamic Capabilities Mediator and Open Innovation Moderator

Thanh Tiep Le and Abhishek Behl

***Abstract*—The uncertain and rapidly changing environment raises deep concerns about supply chain resilience (SCRe). Particularly in the small and medium enterprises (SMEs) settings, the lack of resources of SMEs becomes more pressure when considering SCRe. Although the literature on supply chain management and SCRe is increasing, understanding how artificial intelligence (AI) adoption can contribute value to SCRe needs further study. In this article, we aim to bridge the gap by examining an integrated framework of factors conceptually assumed to enable SCRe improvement. Grounded in information processing theory and dynamic capabilities view perspective, conceptually, AI has a relationship with SCRe; simultaneously, dynamic capabilities (DCs) mediate this relationship, and open innovation (OI) moderates the influence of AI on DC. The findings of this study suggest that, in the face of disruption and an uncertain environment, AI is a strategic tool for firms to improve DC and, ultimately, SCRe. Additionally, OI plays a moderation role in the impact of AI on DC. Drawn upon the knowledge-practice gap in this field, this study is among the few currently providing empirical evidence on optimizing AI benefits for building lasting SCRe based on DC development.**

***Index Terms*—Artificial intelligence (AI), dynamic capabilities (DCs), open innovation (OI), supply chain resilience (SCRe).**

I. Introduction

SUPPLY chains (SCs) across the globe have experienced a severe crisis due to the COVID-19 pandemic [1], [2]. The closure of borders and social distancing due to COVID-19 has caused severe SC disruption [3] and unprecedented socioeconomic crises [2]. Consequently, it threatens business continuity and supply chain resilience (SCRe) [2], [4]. Furthermore, in this context, small- and medium-sized enterprises (SMEs) are significantly affected because they represent the hardest hit sectors [5]. Likewise, Kindström et al. [6] press that SMEs are most threatened by the COVID-19 crisis due to their relative vulnerability. On the other hand, SMEs face resource constrained in various ways [7]. While in the context of crisis, a substantial investment is required to achieve resilience [8]. In this regard, developing their capabilities in a more challenging era may necessitate optimizing the external resources as a way to overcome the resource constraint [7], especially in the context of a crisis.

Furthermore, taking an approach from another perspective, SMEs represent more than 90% of all businesses worldwide, making them the backbone of the multisector world economy [5]. Based on this situation, SMEs should be focused on further research on SCRe. Specifically, how SMEs can enhance their resilience in the new era. Considering further the context of SMEs, resilience depends on many factors, among which information systems play a significant role [8]. In addition, the lack of a proactive and preventive strategy has weakened SMEs' resilience [9]. In this context, artificial intelligence (AI) is proposed as a mechanism that can drive SCRe in a way that promotes business continuity capabilities [10]. It connects stakeholders in real time [10]. According to Ivanov et al. [11], AI is one of the emerging technologies capable of processing and analyzing information based on big data with high prospects for SC risk analysis, thereby enabling SCRe. SC disruptions adversely affect not only businesses but also the economy at large. This impact is severe during global crises, such as the COVID-19 pandemic [12]. Therefore, understanding how to build SCRe becomes very important because of its crucial role not only for the businesses' survival but also the value chain sustainability [13] and the economy development at large

The current literature on the strand of SC management has explored a wide range of AI-derived benefits [3]. For instance, AI powers desired visibility and transparency, leading to greater flexibility in the value chain [14], better management of inventory and reduction of fraud [15]; enables SC security and traceability, which, in turn, creates superior performance and better trust from customers and partners for the business [16] and decision-making efficiency [17]; and boosts SMEs' dynamic capabilities (DCs) [18]. Despite this, challenges exist around concerns about whether, when, and how businesses can benefit from adopting AI [19]. A part of that transitioning from AI-powered insights to actions primarily relies on decision-making capacity; in this regard, the normal practice of decision makers often comes back to their intuition and experience because they feel more comfortable doing so rather than informed insights [19]. Furthermore, other challenges may affect the extent to which businesses benefit from AI, for instance, behavioral and cultural perspective [19]; capabilities

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[20]; and knowledge and infrastructure [21], [22]. In the field of SCRe, understanding inclusively the circumstance in which AI contributes to SCRe needs further study [1], [3].

Furthermore, would AI produce such benefits in any circumstance? This question arose because, as highlighted above, challenges still exist that may prevent businesses from benefiting from AI. Through this lens, it is assumed that it is necessary to employ other elements in the AI–SCRe relation to be able to make the optimal use of AIs advantages in improving information processing efficiency to deal with risks, uncertainties, and the environment dynamism [23], [24]. Furthermore, not all businesses can get insights from adopting AI and use it for efficient operations [25]. The above arguments suggest that there is a need for further research on how SMEs can enhance their resilience, given the resource constraints that they face in various forms. However, current literature on SCRe was scant [7].

In response, this study aimed to bridge the aforementioned void by examining the circumstances in which AI adoption is assumed to affect SCRe with the involvement of DC as a mediator and open innovation (OI) as a moderator. This frame is operationalized through the lens of organizational information processing theory (IPT) and dynamic capabilities view (DCV). From the perspective of IPT and DCV theories, practicing OI may help firms enrich insights through promoting exchanges with their external resources. As a result, such practices may enable firms to enhance their DC in a variety of ways. Grounded in the IPT point of view, assuming that the extent to which businesses implement OI can influence the level of AIs impact on DC. Given this argument, assuming that OI is highly relevant to moderate the link between AI and DC. Likewise, grounded in the DCV perspective, assuming that DC can boost connecting stakeholders on an information-based and data-driven basis, assuming that it is highly relevant to mediate the relationship between AI and SCRe. In SMEs' settings, individual factors (experience, knowledge, and various social relationships) prevail over other factors, such as information processing systems and state of capabilities in forming decision-maker judgments [26]. This approach has raised whether this practice is widely applicable to SMEs in every context. Therefore, understanding whether IPT and DCV can operationalize the frame of AI adoption, SCRe, DC, and OI in the SME setting is vital. Through this lens, the main objective of this study is to explore two significant aspects. First, the mechanism by which SMEs in the SC in emerging economies can build SCRe. Second, whether or not AI adoption can contribute to building SCRe directly and through DC and OI in the new settings. Accordingly, this study aims to address the major questions systematically as follows.

RQ1: To what extent does AI contribute to SCRe for SMEs in emerging economies?
RQ2: Whether DCs play a mediating role in the link between AI and SCRe?
RQ3: Whether OI play a moderating role in the impact of AI on DCs?

Although a considerable number of studies have emerged in the recent past, the contribution of this study is timely and significant, particularly to SMEs. This study is distinctive for providing insights and an empirical-based mechanism for optimizing AIs benefits to advance SCRe through a theoretical lens of IPT and DCV. This study contributes to expanding the knowledge of the current literature on digital technologies (AI) and SC management in several ways. First, this study contributes to a further understanding of the specific part of technological advancement, which can be applied to enhance SCRe. Specifically, this study provides insights into how SMEs in emerging economies use AI to build and develop SCRe. Additionally, it provides a profound understanding of how integrated OI and DC enable firms to address resource constraints that are inherently considered the major issue of SMEs. Particularly, unlike other studies in which DC was conceptualized as a part of SCRe, this study theorizes DC as separate from SCRe. From this approach, DC is viewed as a requisite of SCRe and SCRe as a state of SCs health. Second, this study contributes to expanding IPT and DCV by enhancing the theoretical understanding of how AI adoption contributes to SCRe in the context of SMEs in emerging economies. Furthermore, this theoretical ground is crucial for a better understanding of how DC and OI engage in the link between AI adoption and SCRe. Third, this study provides a novel approach to boost SCRe for SMEs by utilizing internal–external resources, simultaneously forming and enhancing DC, as an optimal way to transform AI resources into value. Drawing on the findings of this study, knowledge and practical implications are provided to professionals, academics, and other stakeholders concerned with SC management and its efficiency. Especially in the context of uncertainties and disruptions, SCRe is critical for its viability and the socioeconomy recovery of the nation and the region. The originality of this study is the provision of empirical evidence on the mechanism by which SMEs can derive value from AI for SCRe. This contribution is especially significant to SMEs because their dominant characteristic is a lack of resources. OI and DC can help SMEs overcome this constraint in this scenario to adopt AI for SCRe. This study's findings may contribute to not only broadening the boundaries of DCV from within a company to the SC as a whole but also enriching the existing empirical studies on how to build and enhance SCRe, especially for SMEs in emerging economies. This study, therefore, contributes to the current literature on operations management and SCRe as a fundamental insight for both theory and practice and for future research.

The rest of this article is organized as follows. Section II is dedicated to underpinning theories and conceptualization. Section III describes the research framework and hypothesis development. Section IV describes the research methodology. Section V describes the result interpretation, and Section VI presents the discussion. Finally, Section VII concludes this article.

II. Conceptual Background

A. Underpinning Theory

This study examines the connection among AI, DCs, SCRe, and OI from the lens of organizational IPT and DCV perspectives. According to Juan and Li [27], DCs are comprehensively viewed, including resource-based and knowledge-based DC and SCRe. In this regard, resource-based DCs are fostered

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by knowledge-based DC that require access to information to support sense opportunities, recognize them, and seize and transform them into reality [27]. Furthermore, knowledge-based DC can promote collaboration between SC stakeholders, which in turn grows the collective intelligence of the SC network through enhanced sharing of information and knowledge among stakeholders [27]. From this view, IPT is theoretically related to DCV in the sense that IPT enables access to information and knowledge to facilitate DC. Given the characteristic constraints of SMEs in emerging economies (resource constraints) [28], an understanding of how to utilize internal and external resources through OI for DC is critical for SMEs.

From the IPT perspective, information processing is critical to businesses for their viability and sustaining operational efficiency, particularly in the face of increased uncertainty. In addition, businesses necessitate developing capabilities enablers and having processing information capabilities to manage SC disruptions [23]. In practice, processing information to deal with risks, uncertainties, and dynamism requires high organization capabilities [23], [24]. Therefore, from the IPT perspective, AI is viewed as a static capability that enables organizations to improve their DC and drive SCRe. AI adoption and SMEs' resilience in the SC were explored in the Vietnamese settings by Dey et al. [29]. In this regard, SCs firms' resilience can be supported by leveraging AI to deliver appropriate responses to unprecedented disruptions through data-driven decision making [29]. However, the association between AI adoption and SMEs' DC for SCRe has not been focused in the recent literature.

Along with IPT, this research integrates DCV as a new approach on the theoretical ground to examine how DC plays a distinct role, mediator, in the link between AI adoption and SCRe. DCV is known as an extension of resource-based view (RBV) theory [30], [31] that can participate in explaining how an enterprise's existing resources can be evolved to facilitate a transition from static-based capabilities to dynamic-based capabilities and an uncertain environment [32]. In this respect, while the RBV theory intensively prioritizes the selection from existing resources, DCV focuses on developing, acquiring, and exploiting resources [33], [34]. Current literature supports the assertion of the usefulness of DCV in orienting enterprise resources to enhance capabilities to gain competitive advantage [10], [35]. In a dynamic environment where data and information are crucial to an enterprise's growth strategy, especially in an uncertain environment, DCV enables organizational alignment to leverage big data systematically [36]. Applying DCV in this framework, AI is viewed as the basis of resource capabilities that form a ground for building higher order capabilities, such as SC collaboration and adaptability [37]. From the perspective of DCV, higher order DCs are critical for enhancing low-order DC for value creation coupled with improving organization's problem-solving capabilities that enable businesses to initiate change in their environment [38]. In the context of emerging economic SMEs, resources and capabilities constraints hinder them in survival and development, especially in the context of an unpredictable changing environment [28]. In this context, understanding how to leverage internal and external resources is crucial for SMEs to promote multidimensional DC for SCRe in the environment where they operate.

B. AI Adoption

AI adoption is conceptually perceived as a field of computer science that includes developing systems capable of performing tasks that normally require human intelligence [39]. According to Grover et al. [40], AI is viewed as the ability of a system to acquire knowledge by analyzing the external environment's data and using the acquired knowledge to adjust or plan new strategies against environmental changes. In application, although AI is typed depending on different forms, within this research framework, the conceptualization of AI is focused on its ability to acquire knowledge and learn to enhance resilience to disruptions [41]. With its distinctive design, AI can overcome the limitations of human cognitive information processing and process significant amounts of data to support effective and efficient decision making [42]. At the same time, AI enables users to generate new ideas to facilitate human–machine interaction during problem analysis [22]. However, there is still little understanding regarding how AI can build capabilities at the SC level.

In the SMEs' settings in emerging economies, there is evidence of a gap in their implementation of AI solutions [43]. Such a gap exists due to the lack of resources needed (financial and nonfinancial) to implement AI [43], for instance, lack of knowledge, skilled human resources, data culture, investment and commitment to maintaining investment, etc., and lack of evidence and little visibility of return on investment [43]. Consequently, these constraints hinder the realization of AI in business operations. Additionally, there is a piece of evidence that AI diffusion varies at various levels (country, sector, and firm size). In this regard, it is assumed that businesses that lack internal resources to implement AI may be compensated by leveraging external resources [18]. Overall, the existing literature on the AI strand highlights various benefits of AI for operational efficiency; however, the integration of factors to reduce these constraints on the influence of AI on SC management efficiency (SCRe) has not been sufficiently explored.

C. Open Innovation

OI is a commonly used concept in recent decades that refers to a state where a firm utilizes internal and external resources for innovation processes [44]. OI is a step forward in perception and practice by shifting from traditional patterns (closed innovation) to OI to optimize an entity's internal and external resources in response to a changing environment. OI and its relevance to AI are receiving high attention from stakeholders as a means of helping businesses, especially SMEs, and improve their operational efficiency in the digital economy era [45]. From a DCV point of view, OI enables one to exchange knowledge, information, data, etc., with one another internally and externally [46], which, in turn, is assumed to be able to address the constraints caused by a lack of knowledge, data culture, and necessary resources for businesses to benefit from AI. Because the dominant characters of AI provide not only a method of achieving productivity gains but also a fundamental innovation of the tools SMEs can use to innovate [47].

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D. Dynamic Capabilities

DCs were developed as capabilities to innovate competencies in response to changing environment [31]. In this regard, DC enables businesses to shape, grow, and reconfigure resources and competencies oriented toward building and retaining resilience [31]. DC enables a business to build and renew resources and assets in ways that benefit both inside and outside of its boundaries, reconfiguring them as needed to innovate and respond to changing environment [31]. Chari et al. [48] emphasized that the firm's sustainability is influenced by its DC (skills, competencies, and resources), which, in turn, causes an impact on its entire SC efficiency. In the area of SCRe, El Baz and Ruel [49] state that realizing SCRe requires the capability to leverage resources, capabilities, and processes. In this respect, DC is assumed highly relevant to realize SCRe. Especially for SMEs in emerging economies, which are known for their limited resources to make changes as needed, multifaceted DC enables leveraging higher and lower order DC assumed to be able to help SMEs proactively respond to unpredictable fluctuations.

In an increasingly competitive environment, firms should reconfigure their resources and capabilities to stay competitive and adapt to changes in the uncertain environment [48]. From a DCV perspective, DC should be viewed as the main source of value creation that allows businesses to identify opportunities and threats, which, in turn, facilitates informed decision making as to whether to exploit or neutralize them based on the firm's resources and capabilities [50], [51]. In the context of uncertainties, DC-derived values are associated with innovations, strengthening SC for adaptability and recovery [52]. In the SME settings, DC can help businesses overcome obstacles caused by a lack of resources (financial related) to invest in research and development activities and advanced technologies by enabling environment scans to understand the movement and opportunities in the marketplace [53]. However, despite the importance of DC, especially for SMEs, the current literature shows a lack of consensus that requires further context-based studies [54], [55].

Another argument in the study of Laaksonen and Peltoniemi [56] argues that DC does not necessarily lead to better performance, but the extent to which DC contributes to the firm depends on the level of the firm's ordinary capabilities and the environment dynamism [57], [58]. Given the above discussion and argument, DC should be explored independently and its measures should be multidimensional to fit this research setting (SMEs that often face limited resources and managerial issues that primarily focus on ordinary capabilities to make a living in short view). In this context, in addition to direct DC measures of sensing, seizing, and reconfiguring, DC measures of staying connected in the SC and network; and resolve conflicts among its stakeholders, are also crucial because they are assumed to facilitate sensing, seizing, and reconfiguring DC.

E. SC Resilience

SCRe carries inconsistent conceptualization in the current literature. For instance, SCRe is considered one-dimensional (1-D) DC in most previous studies [59] or multidimensional DC [60]. Different dimension requires different types of resources and capabilities of the firm [59]. In the context of this study, the target population is SMEs, which are known to have limited resources [28]. Therefore, it is necessary to consider DC as a distinct factor and not embedded in SCRe. In this respect, SCRe is viewed as the health condition of the SC, a state in which SC continuity is controlled and SC operations are well performed as "normal" regardless of changing environment. Compared with other studies, where SCRe is considered a unidimensional capability or a complex DC [59], this study conceptualizes SCRe as a state in which the SC is comprehensively equipped to overcome unpredictable fluctuations and ensure its continuity in the environment. From this approach, DC is viewed as a distinct factor and a requisite of SCRe for SC viability.

In the context of the COVID-19 outbreaks, SC is severely disrupted; therefore, SCRe is highly needed [61]. Meanwhile, backup plans are essential in the face of SC disruptions; importantly, such plans require demonstrating appropriate resilience [62]. Folke et al. [63] emphasize that SCRe is characterized by persistence, adaptability, and transformative readiness. At the same time, Brandon-Jones et al. [64] stress that SCRe depends on the ability to respond to and recover from disruptions. In the SME setting, their SCRe depends on many factors. According to the authors in [8] and [65], such factors are related to financial and nonfinancial areas (teamwork and culture, capacity, information system, and financial potential). However, SMEs characteristically face resource constraints in response to a rapidly changing environment. In addition, their lack of proactive strategy has weakened their SCRe. In this context, therefore, understanding how to drive SCRe for SMEs in the age of technology is a very important and exciting area due to its importance to the overall economic recovery and growth [7].

III. Research Framework and Hypotheses Development

SCRe in the face of unpredictable disruptions requires enterprise-timely DCs. However, DC may not accommodate such uncertainties in a timely manner without the support of the AI platform. In this context, AI is viewed as an enterprise's tool for processing information whose value is beneficial to analyzing rich data and providing useful knowledge to the firm [4], [10]. Particularly for SMEs in emerging economies, this framework is assumed to be meaningful in addressing the widely known constraints of SMEs (financial and nonfinancial, knowledge, and capabilities included). SMEs dominate the national economies in the globe in terms of numbers, job creation, and contribution to GDP [28]. In Vietnam, the number of small and smaller enterprises accounts for the majority of SMEs. They face more resource constraints than larger scaled enterprises [28]. On the contrary, smaller SMEs contribute less value to the value chain than larger SMEs. Based on their resource constraints, providing an understanding of how to leverage internal and external resources simultaneously and understanding under what circumstance AI can boost their DC for SCRe is assumed to be significant not only to SMEs SC but also to society at large. Drawn upon the above arguments, this study focuses on

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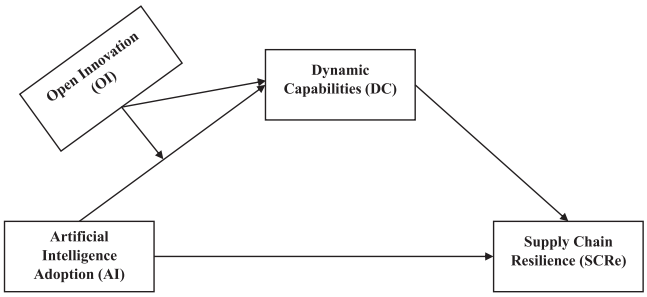

Fig. 1. Theoretical model.

empirical research on the relationship among AI, DC, SCRe, and OI from the lens of IPT and DCV perspectives. Fig. 1 illustrates the proposed research model in the present study.

A. Connection Between AI and SCRe

In the SC field, Sullivan and Wamba [41] found that AI use has a positive impact on firm resilience to SC disruption. AI adoption is viewed as organizational agility capacity in the digital age that enables quick and smart sensing of supply and demand conditions in real time. In turn, businesses can quickly sense, seize, and transform emerging opportunities and mitigate their risks in a volatile environment [25]. In the settings of SMEs, AI can benefit firms by providing dynamics that facilitate a business environment and conditions, business model innovation and practices, etc., to optimize adaptability for viability and growth [43]. In SC management, particularly in an uncertain and disruptive environment, AI plays a crucial role in facilitating SCRe [41]. Specifically, AI helps businesses shape their resilience to SC disruptions [41] in a way that leverages information processing and embeds it in the broader organizational processes, shortening SC response and recovery times [10].

In this research scenario, the role of AI in driving SCRe can be further explained through the theoretical lens of DCV and IPT perspectives. From the IPT perspective, firms are viewed as open social systems seeking to execute their business strategies to mitigate uncertainties during decision making [66] by acquiring knowledge and learning from information and data from the environment [67]. In the proposed framework of this study, AI focuses on the aspects of infrastructure and skilled resources to apply AI-based information processing systems; forecasting and predicting; developing DCs; promoting learning; advancing the decision-making process; and engaging all levels in the broad process of the firm conceptually promote SCRe in a way that enhances SC adaptability, responsiveness to the changes due to environmental uncertainties, and recovery from disruptions [4], [10]. Naz et al. [68] highlight that AI is an enabler of SCRe. In the present research settings, the SME community, the tendency to join business associations increases because they know that cooperation and collaboration are vital to building the resilience of the value chain and reducing the risk of disruption. Therefore, it is assumed that adopting AI can foster stronger cooperation and collaboration through information and knowledge sharing among SC network members, facilitating the resilience of the SC to disruptions. Drawing from the above discussion, the connection between AI and SCRe can be proposed as follows.

Hypothesis 1 *(H1) AI positively and significantly affects SCRe.*

B. Impact of AI on DC

The above discussion and arguments indicate the high prospects of AI for the development of capabilities through its ability to acquire knowledge by analyzing data from the external environment and using the acquired knowledge [40]. Grounded in IPT, a firm is viewed as an information processing system that is capable to mitigate the adverse effects caused by uncertainties by improving its capability to collect and process information from the environment [67], [69], [70]. In this context, AI sharpens the firm's information processing ability, motivating them to improve DC. Adopted from Lee and Rha [71], DC is measured through the ability to feel the movement of the business environment; observe and judge changes well in the SC; stay well connected; rapidly resolve conflicts; make timely decisions; restructure resources in a timely manner; reconfigure resources successfully; and create novel combinations of resources. Therefore, it is assumed that a firm that applies AI can develop DC to identify better and capture rapidly changing market opportunities and adapt its behavior to market needs, thereby responding to market changes [65]. Drawing on the above discussion, the impact of AI on DC can be hypothesized as follows.

Hypothesis 2 *(H2) AI positively and significantly affects DC.*

1) Relationship Between DC and SCRe: Amid uncertainties and unpredictable movements of the COVID-19 pandemic, Chowdhury and Quaddus [72] emphasize that the ability that respond quickly, make quick decisions, and reconfigure the resource base significantly affects the extent to which the business copes with the risk of disruption. Such capabilities are needed for firms to survive during the pandemic, and strengthening capabilities is imperative to build a firm's resilience [72], [73]. Among the observable factors of DC, the capability of reconfiguring the resource base has the strongest effect on SCRe [72]. From the DCV perspective, DC is viewed as playing a decisive role in driving SCRe, thanks to its dynamism, flexibility, and agility to make timely appropriate decisions and actions to drive resilience to disruptions. Furthermore, in SCRe, the extent to which the company can sense, learn, integrate, and coordinate to be resilient to disruption is critical [71]. Given the above discussion, the relationship between DC and SCRe can be hypothesized as follows.

Hypothesis 3 *(H3a)* DC has a positive and significant relationship with SCRe.

2) Mediation Role of DC in the Relationship Between AI and SCRe: Belhadi et al. [4] found that AI ultimately impacts SCRe and SC performance. In this context, AI leverages SCRe to enable businesses to constantly learn from market data to understand what is happening and to base decision making in response to changing environments during disruptive events [40], [74]. From the IPT perspective, in the face of uncertainties,

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businesses are always in a state of information processing to make the most appropriate decisions and undertake the most appropriate changes to respond promptly to the rapidly changing environment. In this context, AI advances this process through its capabilities to aggregate and process data and information on a big base. In addition, from the DCV point of view, AI is viewed as a tacit capability underpinning DC development oriented toward capabilities to sense opportunities and threats, seize opportunities, and reconfigure the resource base to respond to uncertainties and recover from disruptions. As emphasized by the authors in [75] and [76] focusing merely on AI development does not solve external shocks, but importantly, assessing the extent by which enterprises can absorb insights derived from AI and SCs is critical for SCRe, from the IPT and DCV theoretical perspectives. This argument assumes that DC may be relevant in addressing this constraint.

An approach by Lee and Rha [71] argues that building DC-SCs is essential for resilience through mitigating the negative effects of SC disruptions while proactively creating opportunities to grow, seize, and transform them into reality. On the one hand, resource-based consideration and DCs with subcapabilities (sense opportunities, seize, and transform opportunities into reality) help business growth. On the other hand, DCs and knowledge-based consideration can promote collaboration among SC stakeholders and, thus, facilitate SCRe. In this setting, DCs encompass the ability to feel the business environment movement; observe and judge changes well in the SC; be connected; address conflicts quickly; deliver decisions timely; reconfigure resources timely; and allocate resources in new ways to appropriately address new coming missions [71], which enable SMEs optimally use of AI advantage to advance SCRe. In SMEs settings, DC is considered a mainstream resource for value creation by enabling the business to identify market movements (opportunities, threats, and other related), facilitating efficient decision making [51], [77]. Based on the above arguments, the role of DC in the relationship between AI and SCRe can be hypothesized as follows.

Hypothesis 4 *(H4)* DC mediates the relationship between AI and SCRe.

3) Moderation Role of OI in the Influence of AI on DC: It is stressed that if there is a knowledge gap in adopting AI, then SCRe is likely to be impacted. In SME settings, openness enables firms to leverage external expertise and solutions for AI adoption to compensate for their lack of internal capacity [43]. This notion is supported by Drydakis [18], who argues that SMEs with perceived resource constraints, thus requiring them to adopt AI, should be open to accessing external knowledge and, simultaneously, identifying the existing knowledge of the company. It is emphasized [65] that AI positively correlates with information obtained from the external environment. In this context, OI is characterized by encouraging finding and exchanging knowledge between internal and external sources; advancing collaboration and cooperation between stakeholders in the SC to achieve better resilience; enabling data sharing amongst stakeholders to enrich the database; innovative products and services; accessing new technology; and combining inside-out and outside-in processes, and can associate with AI to develop DC. According to Teece et al. [31], DC is a firm's ability to integrate, develop, and reconfigure internal and external competencies to address rapidly changing environments.

From the DCV perspective, OI can promote DC development in a way that enables one to access multiple sources of knowledge and information, which, in turn, helps one enhance their DC. From an IPT point of view, AI facilitates one's information and data processing; in this context, OI can associate with AI in a way that utilizes acquired knowledge and information to generate the optimal outputs of information processing to help businesses enhance their DC toward enhancing their ability to sense opportunities and threats, to seize opportunities, to reconfigure the resource base to respond to rapidly changing environments, and recover from disruptions. In the context of this study, the form of participation in business associations of SMEs can promote OI, at least from enterprises in the same association. Because these businesses may be more open to each other and, therefore, more willing to share, in this regard, as OI evolves, it will better assist AI in performing its role and function, leading to better development of DC. Based on the above arguments, the role of OI in the relationship between AI and DC can be assumed as follows.

Hypothesis 5 *(H5)* OI moderates the influence of AI on DC.

IV. Research Methodology

A. Survey Instrument Development

A structured questionnaire was developed for the survey to collect primary data to measure construct in this framework. The questionnaire was structured with two main sections. The first section presented the basic information of the respondents, and the second section presented the questions for collecting data. In the present research, the questionnaire was developed based on a five-point Likert scale whose extreme points indicated a range of responses to the corresponding questions. Specifically, one indicated "strongly disagree/disagree," while five indicated "strongly agree/agree." The respondents were asked to answer by choosing the score closest to their agreement with the respective statement.

The questionnaire was verified through the necessary steps to ensure its appropriateness to optimize the possibility of collecting the most accurate and sufficient data. Accordingly, the questionnaire was reviewed by academics and professionals in the field. This assessment focused on the appropriateness of constructs, measures, nonduplication, and clarity. After this step, the questionnaire was improved where needed to be ready for a pilot test by participants with similar backgrounds to the target population. There were 56 people who participated in this pilot testing. This step resulted in a complete questionnaire for the official survey. Based on the respondent's feedback, the points that need to be improved were not significant, mainly related to the questionnaire's layout. For instance, some respondents suggest adding two to three more sentences to describe the survey further and state its purpose. In the present research, these constructs were designed as reflective constructs.

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B. Sampling Design and Data Collection

The empirical context of the present study is the firms that adopted AI techniques in their SC operations. The analysis unit of this study was the focal business from which the most involved people were invited to participate in the survey. In the context of this study, the sample population focuses on SMEs operating in economic settings in Vietnam for some reasons. First, at the microlevel, SMEs in Vietnam appeared to be innovative; however, at the current status of resources' constraints, their innovations are mainly focused on improving products and processes to increase competition (cost conscious) in the market [78] and strategically innovate toward building resilience to disruptions caused by environmental shocks. Second, at the macrolevel, the national innovation system is still at an early stage of development, and the link in innovation is still weak, considering the Vietnamese context [78]. However, Vietnam is an emerging economy known for its dynamism in socioeconomic development [79]. Moreover, the Vietnamese economy is characterized by the majority of SMEs recognized for their significant contribution to socioeconomic development. Based on the above highlights, an assumption arose that research using the context of SMEs in Vietnam as an empirical area may be generalized to other similar contexts.

The sampling process was conducted conveniently through business associations that help businesses develop their business activities. As a process, an open letter was first sent to the management board of the associations of SMEs to succinctly present this research and its potential implications for the future of SMEs, especially in uncertain and unpredictable contexts recently. Thereby, the researcher expressed his intention to survey the enterprises in the associations if willing. Through the internal process of the SME associations, a list of businesses that agreed to participate in the survey was prepared, noting that it is for only this research. This process resulted in a total of 706 samples that were used for the survey. The survey was conducted using an online platform from May 2022 to August 2022. In order to motivate respondents to answer questions as sufficiently and accurately as possible, a cover letter was sent along with the questionnaire. In this way, such a cover letter outlines this research and its potential contributions to addressing critical concerns in the field of the SC. End of the survey, there were 496 responses obtained (accounting for a 70.25% response rate) that were then screened to select the eligible samples; as a result, there were 477 valid samples remaining, accounting for a 67.56% response rate. Table I summarizes the characters of valid samples. The data collected were primarily analyzed for the analysis sections.

C. Measurement Construct

The measuring items for each construct were developed based on the existing literature and adapted to the current research context. Accordingly, AI adoption includes six items developed based on adopting items from previous studies [4], [10]. In addition, the DCs construct includes eight items developed based on Lee and Rha [71]. Notably, the OI measurement instrument, which includes six items, was developed based on the way in which researchers (academic) and professionals (industrial practice) collaborated to maximize its content validity. In this regard, the relevant literature was based on [80], [81], [82], and [83]. Finally, SCRe was developed based on adopting [84], [85], and [86] and adapted to the research context.

TABLE I
CHARACTERISTICS OF THE OBTAINED SAMPLES

Parameters	Description	N = 477
Gender	Male	219
	Female	258
Managerial position	Chief executive officer	125
	Managing director	156
	Supply chain operation director	196
Managerial experience (years)	Below 5	75
	5 - < 10	89
	10 - < 15	115
	Above 15	198
Size	Small	368
	Medium	109
Sector	Fast-moving consumer goods	116
	Agricultural processed food	259
	Pharmaceuticals	102
History (years)	Below 5	47
	5 - < 10	78
	10 - < 15	145
	Above 15	207

D. Data Analysis

This study employed the partial least square structural equation model technique to analyze hypothesized relationships. According to Dubey et al. [10], the structural equation model (SEM) is a suitable approach to examine the interrelationships between the independent variable(s) and the dependent variable(s) because it includes a set of regression analyses. Moreover, SEM is proper for the context in which the role of a variable is mutable for regression analysis. In this regard, a dependent variable can become an independent variable of another.

V. Data Analysis and Discussion of Results

A. Characteristics of Obtained Sample

The collected samples have statistical characteristics, as shown in Table I. The size of the enterprise is determined based on the criteria of [78], considering the number of employees of the enterprise. Accordingly, enterprises with employees from 10 to 49 are determined as small scale, and from 50 to 249 are classified as medium scale. In addition to the company size statistics, the remaining statistical characteristics may reflect the desired population in the study context.

B. Measurement Model Validation

Prior to presenting confirmatory factor analysis (CFA), correlation, itemwise correlation, and factor analysis are discussed.

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TABLE II
CFA RESULTS

Variables	Items	AVE	Min	Max	Mean	Standard Deviation	Construct level correlation			
							AI	DC	OI	SCRe
AI	AI1	0.646	1.33	5	3.5	0.73304	1	0.734	0.181	0.677
	AI2									
	AI3									
	AI4									
	AI5									
	AI6									
DC	DC1	0.623	1.38	4.88	3.5671	0.71373	0.734	1	0.183	0.72
	DC2									
	DC3									
	DC4									
	DC5									
	DC6									
	DC7									
	DC8									
OI	OI1	0.58	1.5	5	3.6792	0.68007	0.181	0.183	1	0.042
	OI2									
	OI3									
	OI4									
	OI5									
	OI6									
SCRe	SCRe1	0.665	1	5	3.6876	0.67576	0.677	0.72	0.042	1
	SCRe2									
	SCRe3									
	SCRe4									
	SCRe5									
	SCRe6									
	SCRe7									
	SCRe8									

Importantly, the normality distribution of data was evaluated by Skewness and Kurtosis analysis. Accordingly, the highest absolute value of Skewness for all measurement items is 0.816 (<3), and the greatest absolute value of Kurtosis is 1.052 (<10). This confirms that this study's normal distribution is not problematic [87]. In addition, the results show that all items have a total correlation greater than 0.3, proving the good fit of the items.

CFA was performed to evaluate construct validity according to the guidance of Fornell and Larcker [88]. In this regard, considering the indices according to the proposed thresholds determines the satisfactory level of aspects of measurement model evaluation (constructs' reliability, convergent, and discriminant validity), including Cronbach Alpha, composite reliability (CR), average variance extract (AVE), factor loadings, and construct correlations [88], [89]. Given the recommendations of the above authors, with a Cronbach Alpha value greater than 0.7 and a total correlation coefficient greater than 0.3, the scale is reliable [88]. Considering the correlation between the constructs, the results, as presented in Table II, show that there is a positive correlation between them. Accordingly, indicating that any change in one will lead to a change in the other one in the same direction. Specifically, the results show that AI has the highest correlation with DC, followed by SCRe. Likewise, DC has a fairly strong correlation with SCRe after that of AI and DC.

In addition, with CR greater than 0.7 and AVE greater than 0.5, the convergent validity is established. Furthermore, the square root of AVE and intercorrelation were evaluated to determine discriminant validity. This study shows that the AVEs square root value (value shown at the top of each column) is greater than their intercorrelation; therefore, discriminant validity was confirmed. Table II presents the CFA results that illustrate the construct validity and convergent validity. Meanwhile, Table III presents the Fornell and Larcker criterion that supports discriminant validity.

Furthermore, the model fit was also evaluated by analyzing the fit indices in accordance with the recommendations of the previous authors. In this regard, indices, such as goodness of fit (GoF) [90], standardized root mean square residual (SRMR) [91], [92], normed fit index (NFI) [88], and R^2 [93], were utilized to assess the level by which the model is determined to be fit. According to Hu and Bentler [90], there are three thresholds of GoF indicating three levels of fit (small, medium, and large). A GoF value greater than 0.36 reflects that the model achieves a large fit. In this study, the obtained GoF value is 0.61, greater than 0.36; therefore, this model is a good fit. Besides, according to [91] and [92], an SRMR smaller than 0.08 indicates a good fit. In this study, the obtained SRMR value is 0.053, less than 0.08; therefore, this model is a good fit. In addition, as suggested by Fornell and Larcker [88], NFI greater than 0.9 indicates a good fit. In this study, NFI has a value of 0.904, greater than 0.9. Furthermore, according to Hair et al. [93], the model has a good fit when R^2 of the dependent variables in the hypothesized relationships achieves greater than 0.1. In this study, DC and SCRe are two dependent variables with R^2 values of 0.552 and 0.565, respectively, greater than 0.1; therefore, this model fits well [93]. Overall, it can be concluded that this model is a good fit.

TABLE III
FORNELL AND LARKER CRITERIA

	AI	DC	OI	SCRe
AI	**0.804**			
DC	0.734	**0.789**		
OI	0.181	0.183	**0.762**	
SCRe	0.677	0.720	0.042	**0.816**

"Bold" values are the square root of AVE values is greater than correlations between the latent variables

TABLE IV
HTMT RESULTS

	AI	DC	OI	SCRe
AI				
DC	0.813			
OI	0.799	0.210		
SCRe	0.143	0.781	0.101	

In addition, heterotrait-monotrait ratio of correlations (HTMT) was used to assess discriminant validity further. Hair et al. [87] state that HTMT smaller than 0.85 supports discriminant validity. This study shows that all HTMT values are less than 0.85, as shown in Table IV. This result confirms that discriminant validity is supported.

C. Common Method Bias (CMB) and Multicollinearity Assessment

CMB and multicollinearity matter were assessed using the variance inflation factor (VIF). According to Kock [94], VIF smaller than 3.0 reflects a state in which multicollinearity is not problematic, and CMB does not threaten the validity of the obtained dataset [95]. In this study, the results show that VIF values are all less than 3.0 (from 1.025 to 2.165); therefore, it can be concluded that multicollinearity and CMB are not significant matters. Furthermore, CMB was further tested using the widely

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known method named Harman's one-factor test. Accordingly, performing exploratory factor analysis and single-factor extraction, resulted in a total variance extracted smaller than 50%, it can be concluded that CMB does not exist. This study's analysis results based on this recommended method generated a total extracted variance value of 39.493 (< 50%). Therefore, it can be concluded that CMB is not problematic in this study [96]. This assessment method has been commonly used in previous studies, such as [25].

D. Structural Model Validation

Using a bootstrapping technique, the structural model was assessed based on the analysis results. In this regard, the table presents path coefficients, t-statistics, p-value, confident intervals (CI 2.5%–97.5%), and variance accounted for (VAF) to analyze the hypothesized relationships between variables in the present model. VAF was used; Fornell and Larcker [88] proposed three threshold values of VAF corresponding to three extents by which the mediator performed. Specifically with a VAF level > 80% (full mediation), 20% $\leq$ VAF $\leq$ 80% (partial mediation), and a VAF < 20% (no mediation).

Given the results, it appears that all hypothesized relationships have a positive and significant effect. Accordingly, the results support the proposed hypotheses on the relationships between variables in this empirical framework, thereby suggesting that AI positively and significantly affects DC ($\beta = 0.740$; $t = 26.626$; and $p < 0.001$) and SCRe ($\beta = 0.323$; $t = 6.709$; and $p < 0.001$). Likewise, DC positively and significantly impacts SCRe ($\beta = 0.483$; $t = 9.829$; and $p < 0.001$). This conclusion, respectively, confirms H1, H2, and H3. In addition, the above results confirm that DC partially mediates the relationship between AI and SCRe ($\beta = 0.357$; $t = 8.991$; 20% $\leq$ VAF = 53% $\leq$ 80%; and $p < 0.001$) that supports H4. The findings indicate that AI adoption has a direct and indirect effect on SCRe through DC. Interestingly, the results show that the direct impact of AI on SCRe is lower than its indirect impact on SCRe ($\beta = 0.323$ versus $\beta = 0.357$). This suggests that the stronger AI adoption leads to the stronger DC and ultimately leads to the higher SCRe. Besides, the results demonstrate that OI moderates the influence of AI on DC ($\beta = 0.108$; $t = 2.590$; and $p < 0.05$), suggesting that the extent to which AI affects DC varies in accordance with OI variance. Given this assertion, H5 is confirmed. Figs. 2 and 3, respectively, illustrate the SEM diagram with and without using a bootstrapping technique.

In this study, the mediating role of DC in the link between AI and SCRe is of interest. Therefore, the mediating effect of DC was further assessed. According to Sobel [97], a mediating role of a variable is established only if it satisfies the following three conditions. First, the independent variable has an impact on the mediating variable. Second, the mediating variable has an impact on the dependent variable. Finally, the participation of a mediating variable will reduce the impact of an independent variable on the dependent variable. Given this theoretical ground, the authors evaluated the mediating role of DC using the Sobel test [98] on statistical package for the social sciences

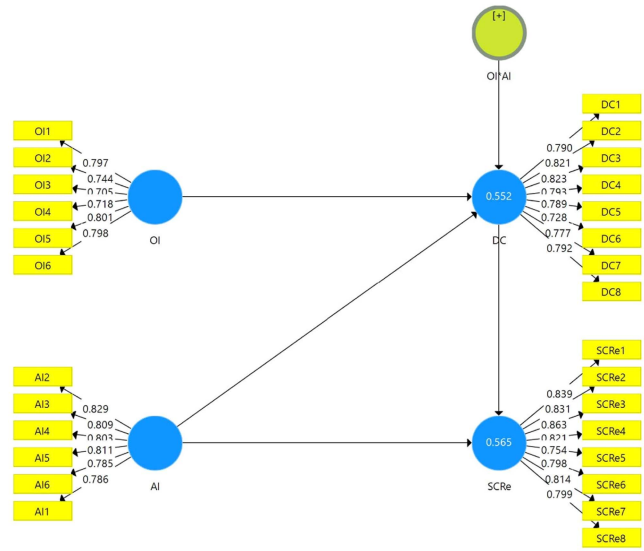

Fig. 2. SEM results without using a bootstrapping technique.

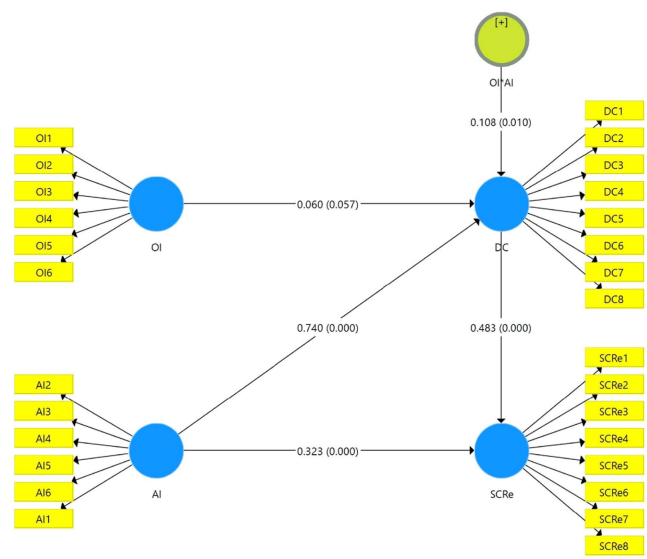

Fig. 3. SEM results with bootstrapping technique.

(SPSS) by manipulating each regression in turn. Therefore, three regressions were performed as follows.

1) *Step 1*: A simple regression was performed to examine the influence of AI on DC. Accordingly, the regression equation is DC = 1.073 +0.713AI. The regression result shows that the Sig value of AI is 0.000 (< 0.05). Therefore, it is concluded that AI (independent variable) has an impact on DC (mediating variable).
2) *Step 2*: A multiple regression for AI and DC on SCRe was performed. The results show that the Sig values of AI and DC are both smaller than 0.05, assuming that DC has an effect on SCRe.
3) *Step 3*: A simple regression was performed to examine the impact of AI on CSRe. The results show that the regression coefficient of AI is 0.623 in step 3, which is larger than the regression coefficient in step 2, which is 0.298, indicating that the participation of DC (mediating variable) reduces

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TABLE V
RESULT OF MEDIATING EFFECT ANALYSIS

	Sobel's (1982) approach		
	Step 1 Dependent variable **DC**	Step 2 Dependent variable **SCRe**	Step 3 Dependent variable **SCRe**
Individual variables			
AI	0.713**	0.298**	0.623**
DC		0.456**	
R^2	0.536	0.564	0.456
Adjusted R^2	0.535	0.562	0.455

TABLE VI
TEST RESULTS FOR THE MODERATED MEDIATION IN THE MODEL

Type of effect	Path	Coefficient	Mean	Standard Deviation	t-statistics	p-value
Direct effect	OI*AI -> DC	0.108	0.101	0.042	2.590	0.010
Indirect effect	OI*AI -> DC -> SCRe	0.052	0.049	0.021	2.487	0.013
Total effect	OI*AI -> DC	0.108	0.101	0.042	2.590	0.010
	OI*AI -> SCRe	0.052	0.049	0.021	2.487	0.013

the impact of the AI (independent variable) on SCRe (dependent variable) [97].

Table V summarizes the results of these regressions. In addition, Table VI presents the test results for the moderated mediation in the present model.

VI. DATA ANALYSIS AND DISCUSSION OF RESULTS

The present study has examined the framework that links AI, DC, OI, and SCRe in the domain of SC in the context of disruptive and unpredictable events. Grounded in IPT, the findings of this study postulate that SC, during disruptive and unforeseen events, such as COVID-19, can be resilient to disruptions by utilizing the information processing capabilities provided by AI. Furthermore, grounded in DCV, our findings postulate that SC can respond to a rapidly changing environment and recover from disruptions through developing AI-based DC and moderated by OI in the context of uncertainties and disruptions. The findings of this study support the findings and arguments of previous studies in some ways. For instance, Chowdhury and Quaddus [72] found that the impact of COVID-19 on the upstream and downstream SC triggers DC development and SCRe improvement, grounded in the DCV perspective. In this respect, the current framework may be viewed as an extended approach compared with that of the study of Chowdhury and Quaddus [72] from two aspects, such as the factor approached as trigger DC and SCRe in the context of SC disruption (COVID-19 impact on upstream and downstream versus AI with consideration of mediation role of OI), and theoretical basis (DCV versus combined IPT and DCV). In addition, Sullivan and Wamba [41] found that applying AI can help improve firm resilience to SC disruptions. In this context, the present study may complement the research framework of Sullivan and Wamba [41] by including the DC factor in mediating the relationship between AI and resilience.

In addition, this study contributes to further elucidation of the current literature on the role of AI in SCRe in several ways. For instance, the study of Modgil et al. [1] explores the impact of AI on SCRe's dimensions. It is found that there are five areas where AI benefits, thereby enhancing SCRe, which are as follows:

1) staying transparent;
2) ensuring door-to-door delivery;
3) providing personalized solutions to SC stakeholders;
4) minimizing the impact of interruptions;
5) facilitating a flexible procurement strategy through the lens of DCs.

However, it took a qualitative approach to a small sample size and did not consider DCs as a single variable in the research model. Moreover, it has not incorporated the organizational IPT perspective as it should have [1]. Beyond this, it has not considered factors that could enable firms to mobilize external and internal resources for necessary innovations. In addition to this, research by the authors in [4] and [25] uses an empirical approach finding in which AI contributes significantly to SCRe for the long-lasting SC through the theoretical lens of organizational IPT. Although, as aforementioned, DCV is considered a theory supporting the IPT assumption, for instance, in the field of AI and SCRe. However, DCV has not been examined along with IPT in their theoretical framework. Beyond that, these studies have not specifically focused on SMEs in emerging economies. Based on these analyses, it shows that the present study's contribution is novel and significant. Given the findings of this study, theoretical implications and managerial implications are suggested as follows.

A. Theoretical Implication

This study provides theoretical contributions in several ways. First, this study contributes to the body of knowledge of the existing literature on the strand of SC management by providing a further understanding of how to foster SCRe in a new approach. In this respect, this study contributes to narrowing the gap in the existing literature on SC management by adding an additional proven pathway to build and enhance SCRe for SMEs in emerging economies. Furthermore, within the given pathway, OI, a factor that is insufficiently explored in the current literature, is employed in the present study and verified to act as a moderator. Additionally, DC is employed as a distinct factor that is verified to act as a mediator in the given mechanism. In a state where the current literature is ambiguous about whether, when, and how AI can contribute to SCRe, this finding provides a new approach that contributes to further elucidation of the above concerns. In addition, SMEs in the research context lack access to SCRe; thus, this finding is a novelty that the present study contributes to the existing knowledge in this field.

Second, this study contributes to the current literature on SCRe by providing a further understanding of how AI adoption can contribute to SCRe. In this respect, through the lens of organizational IPT and DCV, it is explained that SMEs, by adopting AI, have their functions of information processing facilitated, thereby boosting their DC, which ultimately leads to building and strengthening SCRe. This finding may, therefore, enrich the statement of the recent study by Dubey et al. [10], which stated that SCRe can be enhanced by digital adaptability and digital agility by determining AI adoption as the key driver of SCRe, which is also seen as the key driver of digital transformation in organizations in the new era [99]. From this approach, AI adoption is not viewed as high- and low-order DC

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as a part of SCRe as digital adaptability and agility [25], but as a grounding condition to facilitate SMEs to function well with their information processing systems aimed at responding promptly to changes in the environment. In addition, this finding may enrich the findings of a recent study by Zhao et al. [59], which argue that SC digitization has an impact on SCRe's dimensions (absorption, response, recovery) and ultimately affects SC performance by concretizing SCRe's main driver, AI adoption in this mechanism, through the IPT perspective and DCV. In a context where SMEs' access to SCRe is relatively infant and vague, providing a concrete understanding of SCRe's key driver is crucial to putting its application into practice. Furthermore, as emphasized by Dubey et al. [25], the terms "digital adaptability" and "digital agility" themselves still carry inconsistent meanings in the operations management and information systems domain. Therefore, providing a concrete understanding of the driver of SCRe (AI adoption) as in the present study contributes significantly to the knowledge in the SCRe field.

Third, this study adds into DCV by providing a more holistic approach to DC. Specifically, this study provides a new approach to DC through employing DC as a distinct driver of SCRe rather than a part of SCRe. In this regard, through the lens of IPT and DCV, the mediating role of DC in the link between AI and SCRe can be explained because, in addition to the capability to be sensitive, flexible, and adaptive to sudden changes in the environment, still capability that is equally important is to stay well connected in the network and SC at large. This study may, therefore, extend DCV from the perspective of IPT. In this regard, the capability to stay well connected in the network and SC is crucial in terms of up-to-date information. In order to respond properly and promptly to sudden changes in the environment, up-to-date information plays a very important role. This finding may enrich Juan and Li's [27] notion of a holistic view of DC (including resource-based and knowledge-based DCs and SCRe) by providing an approach to DC as a distinct driver of SCRe. In this context, it covers the knowledge-based and resource-based DC; additionally, the capability to stay well connected in the network and SC is fundamental from the IPT perspective. In addition, compared with other studies that consider SCRe a unidimensional capability or a complex DC [59], this study perceives SCRe as SCs strength and SCs health from a multidimensional perspective. From this approach, DC is viewed as a distinct factor and a requisite of SCRe for SC viability.

Fourth, through the theoretical lens of IPT, this study contributes to the existing literature on operations management and SCRe by exploring an additional theoretical ground that supports to explain the phenomenon in this study. IPT enables firms' access to information and knowledge in the external environment and utilizes accessible information and knowledge to drive DC. Current literature on SCRe is rarely explored from the IPT perspective. In this study, IPT was used as an underpinning theoretical ground highlighting the function of the enterprise's information processing system in facilitating DC. Beyond that, this contribution may also contribute to further understanding of the role of digital technologies in enhancing SCRe that is not well understood in theory and application [100], [101], [102].

Fifth, this study contributes to extending IPT and DCV by adding to the existing theoretical ground in the field of operations management and SCRe literature. In this respect, this study adopts the theoretical ground that integrates IPT and DCV perspectives to advance theoretical understanding of how AI adoption contributes to SCRe; and moreover, how DC and OI participate in this linkage. Current research in this area is mainly grounded in either IPT or DCV, or RBV theory rather than combining IPT and DCV. This study, therefore, adds to the existing literature on SCRe by providing a new theoretical approach to theorize the role of AI adoption in facilitating a firm's information processing system function, which, in turn, enhances DC and ultimately leads to building and enhancing SCRe. Furthermore, it can explain how OI is involved in the impact of AI adoption on DC for SCRe. To this extent, this approach can provide a theoretical foundation that enables a further comprehensive analysis of how to build and enhance SCRe.

B. Managerial Implication

Based on the findings, this study proposes several managerial implications as follows. First, to the focal SMEs, this study provides insight to SME managers into the mechanism by which they can use the advantage of AI for SCRe, given the dominant situation of SMEs is the lack of resources. This implication suggests that SME managers should re-evaluate their internal capabilities and current resources needed for AI adoption. This identification is necessary because it helps businesses understand what they are missing, providing an informed basis for businesses to have appropriate strategies and plans to use further external resources. This suggests that although resource constraint is a dominant problem of SMEs that forms a barrier to necessary innovation, SME managers should leverage internal and external resources and utilize AI to form sensing, learning, integrating, and coordinating capabilities for SCRe.

Second, to the other members of the SC, this study implies that other members of the SC should be open to each other so that each can utilize the other's resources because this is necessary for all to overcome the obstacles caused by the lack of resources, as emphasized. Besides resource constraints, data- and information-sharing culture among businesses in the research context may be another obstacle. However, suppose the enterprise is merely based on its internal resources, in that case, it will be difficult to apply AI for SCRe (because of the lack of resources), or it has to be a huge investment while its return may need long time than expected (relevant to the resource constraint). Therefore, this implication suggests that SMEs in the SC should collaborate to optimize each other's resources because this is the most efficient way to overcome the resource constraints for SCR.

Third, to the policymakers, this study implies that openness is an important factor in helping SMEs overcome the limited resources barrier to adopting AI for SCRe. However, in the Vietnamese context, culture-related factors may be another barrier to resource-sharing openness among SMEs because of individual views (competition rather than companion development).

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Therefore, this implication suggests that policymakers should consider supporting conditions, such as policies, institutions, and incentives, to encourage SMEs to share resources for mutual development. This is significant for the survival and development of SMEs in an increasingly competitive and volatile environment and for the long-lasting value chain.

VII. Conclusion, Limitations, and Future Scope of Research

The primary purpose of this study is to explore the connection among AI, DC, OI, and SCRe in the context of SC disruptions. Accordingly, its conceptualization was grounded in IPT and DCV perspectives. The findings of this study provide empirical evidence supporting the proposed framework, simultaneously addressing the research questions posed in Section I. Accordingly, it affirms a connection between AI and SCRe in the context of SMEs in emerging economies. In addition, DC mediates the relationship between AI and SCRe, and OI moderates the impact of AI on DC. Drawn upon these findings, in this article, we operationalized the IPT and DCV perspectives in the proposed framework. This study provided a new approach to the application of AI in SCRe that had received little attention in the current literature. In this respect, this study increased the understanding of the power of AI in developing DC and, ultimately, SCRe. Overall, this study provided significant contributions to theory and practice that may interest academics, professionals, and other stakeholders.

A. Limitations and Scope of Future Research

In addition to the significant contributions as stated, this study has limitations that may be considered for future scope of research. First, this study is based on the primary data collected during a single period when the pandemic gradually came under control in several regions. Therefore, it would necessitate taking larger data at another time to validate the results further. Second, because we do not know how long COVID-19 will last and we cannot predict what will happen in the future, it is necessary to conduct further studies at different periods and possibly employ other factors to extend the empirical analysis framework on how to enhance SCRe. Third, this study did not include control variables. Although control variables are considered unnecessary in this study setting based on discussions with several professionals, they suggest that future research under new settings may include control variables to assess whether they affect focal variable(s) and (if any) then to what extent.

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